

ASL Character Recognition from Transfer Learning

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Abstract— This report explores the application of transfer learning using the VGG19 convolutional neural network (CNN) to classify American Sign Language (ASL) signs. By adapting a pre-trained VGG19 model—originally developed for the ImageNet dataset—the project aimed to leverage its deep learning capabilities to recognize ASL signs through modifications like the inclusion of dropout for regularization, and a softmax output layer tailored for nine specific ASL sign classes. The study involved extensive data preprocessing, including normalization and reshaping, alongside implementing data augmentation techniques such as random flipping and rotations to enhance model robustness. Through a series of experiments focusing on both frozen and unfrozen training phases, the effectiveness of transfer learning was demonstrated, achieving high accuracy in ASL sign classification.

INTRODUCTION

This project focuses on the application of transfer learning using the VGG19 convolutional neural network (CNN) architecture for the classification of American Sign Language (ASL) signs. The integration of deep learning models into ASL recognition systems has been explored extensively in recent literature due to their high accuracy and efficiency in handling complex visual data.

The VGG19 model, developed by Simonyan and Zisserman in 2014, is a deep CNN known for its effectiveness in image recognition tasks. It consists of 19 layers, including 16 convolutional layers, and has shown high performance on the ImageNet dataset [1]. The model's ability to extract high-level features from images made it a valid candidate for transfer learning, where a network trained on a large dataset (ImageNet) can be fine-tuned to a specific, smaller dataset such as ASL signs.

Recent studies have demonstrated the utility of transfer learning for sign language recognition. For instance, Rastgoo et al. (2020) highlighted how transfer learning could significantly reduce the need for extensive labeled datasets while maintaining high accuracy [2]. Furthermore, the work by Garcia and Bruna (2019) on leveraging pre-trained CNNs

for gesture recognition underlines the potential of these models to adapt to different domains effectively [3]. In this project, it was objectivized to leverage the pre-trained VGG19 model, fine-tuning it on a dataset specifically for ASL signs to evaluate its performance in accurately classifying various signs. By adjusting the original architecture and training parameters/specifics, the team sought to optimize the model for our specific use case, balancing between training efficiency and model accuracy.

I. IMPLEMENTATION

Convolutional Layers

Convolutional layers effectively capture the spatial relationships in an image through the application of relevant filters. Each convolutional layer uses a set of learnable filters that have a small receptive field but extend through the full depth of the input volume. For each filter, the convolution operation is performed between the filter and the input, computing the dot product between the filter and the input, and producing a 2D activation map of that filter. As a result, the network learns filters that activate when they see specific types of features at some spatial position in the input.

Data Preparation

The dataset utilized in this project comprises images representing various ASL signs labeled with their respective signs from 'A' to 'I'. The data was initially loaded from .npz files containing the images (data_train.npz) and their labels (labels_train.npz). Each image was transposed and normalized by dividing by 255.0 to convert pixel values into a range between 0 and 1, suitable for neural network processing: The dataset was then split into training, validation, and test sets. A 15% split of the full dataset was reserved for testing, and from the remaining data, 20% was used for validation. This stratification ensures that all classes are evenly represented across each subset. The images were reshaped to fit the input requirements of the VGG19 model, with a shape of (100, 100, 3).

Model Architecture and Training

The VGG19 model was loaded with weights pre-trained on the ImageNet dataset. The model's top layers were not included as the aim was to customize the classifier for ASL signs. A custom neural network was built on top of the base model. This network included a flattening layer followed by dense layers, dropout for regularization, and a dense output layer with SoftMax activation for classifying the 9 classes of ASL signs:

```
inputs = keras.Input(shape=(100, 100, 3))
x = base_model(inputs, training=False)
x_flatten = keras.layers.Flatten()(x)
x_dense_2 = keras.layers.Dense(64, activation='elu', \
                                kernel_regularizer = l1_l2(l1=0.0005, l2=0.0005), \
                                activity_regularizer = l1_l2(l1=0.0005, l2=0.0005))(x_flatten)
x_dropout_2 = keras.layers.Dropout(0.3)(x_dense_2)
outputs = keras.layers.Dense(9, activation='softmax')(x_dropout_2)
model = keras.Model(inputs, outputs)
```

The model was compiled with the Nadam optimizer and Categorical Cross entropy loss function, considering the multi-class, one-hot encoded setup. Training involved two phases: initially training with the base layers frozen, followed by unfreezing all layers for fine-tuning with a very small learning rate to refine the higher-level feature representations.

Handling Unknown Classes

To address the challenge of identifying unknown classes not present in the training data during the contest, a method of outlier detection was implemented. The strategy involved using the SoftMax outputs of the model to determine confidence levels. Instances with maximum SoftMax probabilities below a threshold (e.g., 0.35) were considered unknown. This method leverages the model's inherent uncertainty, which typically manifests as lower confidence for unseen classes.

II. EXPERIMENTS

Experimental Setup

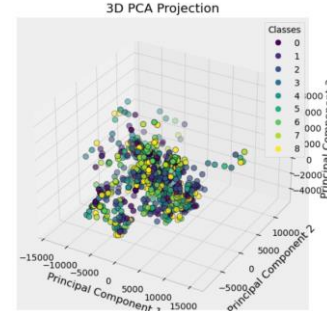
The experiments were structured to evaluate the effectiveness of transfer learning using the VGG19 model to classify ASL signs. The primary goals were to assess the performance of the base model when adapted to ASL sign recognition, understand the impact of data augmentation, and explore the model's ability to generalize from training to unseen data.

Data Visualizations

In this project, several dimensionality reduction techniques were applied to the ASL signs dataset to project it into three-dimensional space, making it easier to visualize and analyze the distribution and separability of the different sign classes. These techniques include Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), Multidimensional Scaling (MDS), and Isometric Mapping (ISOMAP), each projecting the dataset in different ways. For our final implementation, we decided not to use these transformations in our pipeline as our results showed higher performance from the raw data, possibly due to the losses in information that result from some of these transformations. However, these projections still provide a visual aid for analysis of the data.

PCA: Principal Component Analysis (PCA) utilizes orthogonal transformation to convert a set of observations of

possibly correlated variables into a set of values of linearly uncorrelated variables called principal components. The first three PCs were used to project onto a three-dimensional space as they hold the most variance.

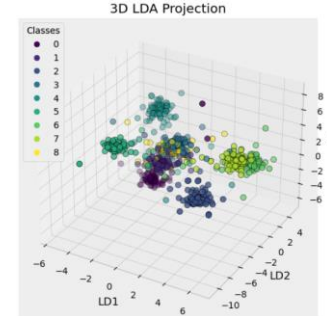


Mathematically, PCA seeks to solve for the eigenvalues and eigenvectors of the dataset's covariance matrix, which are used to transform the high-dimensional data into a new subspace with reduced dimensions:

$$X = X_k = XP_k$$

where P_k consists of the first k principal component vectors.

LDA: Linear Discriminant Analysis (LDA) is a method used to find a linear combination of features that characterizes or separates two or more classes of objects or events. LDA models the differences between the classes of data. It projects the data onto a lower-dimensional space with good class separability to avoid overfitting and reduce computation.

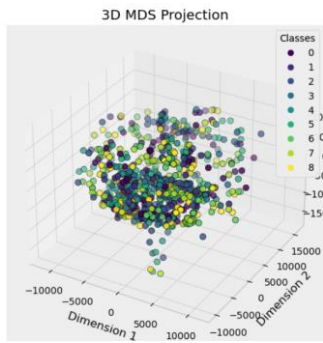


The LDA approach is useful in supervised learning scenarios as it seeks to maximize the ratio of between-class variance to within-class variance in a data set:

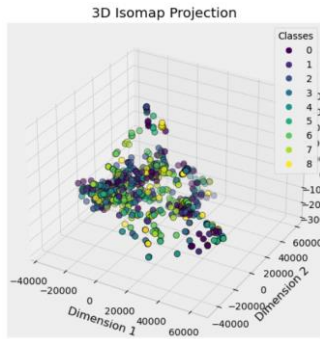
$$\max \frac{S_B}{S_W}$$

where S_B and S_W are the between-class and within-class scatter matrices, respectively.

Multidimensional Scaling (MDS): Multidimensional Scaling (MDS) attempts to model similarity or dissimilarity data as distances in a geometric space. MDS is used in this project to place each data point in three-dimensional space such that the distances between points are preserved as well as possible. It was not very effective due to the high-dimensional data.



Isometric Mapping (ISOMAP): ISOMAP is a nonlinear dimensionality reduction method based on the geodesic distances imposed on a weighted graph. This technique extends MDS by incorporating geodesic distances inferred from the neighborhood graph of data points. It was also not very informative in this case.



Data Augmentation

Data augmentation was experimented with lightly for improving model robustness by artificially increasing the size and diversity of the training dataset. In this project, the following augmentation techniques were implemented using Keras's layer functionalities:

1. Random Horizontal Flipping: Each image has a 50% chance of being flipped horizontally, which is relevant for sign language images where such variation can occur naturally.
2. Random Rotation: Images were rotated by up to 10% of 360 degrees, mimicking slight variations in hand orientation during signing.

The final model did not use data augmentation as it did not significantly improve our model's performance.

Model Training

There were two main processes for training:

1. Initial Training with Frozen Base Layers: The VGG19 base layers were kept frozen to leverage the pre-trained ImageNet features, focusing the training on the newly added top layers. This phase used a higher initial learning rate (0.01) because the training only needed to adjust the new classifier layers.
2. Fine-Tuning with Unfrozen Base Layers: After the initial phase, all layers of the model were unfrozen, and training continued at a very low learning rate (1e-7) to fine-tune the deep features without significant disruption to the already learned representations.

The training process involved monitoring the validation loss with a "ReduceLROnPlateau" callback, reducing the learning rate by 10% if no improvement was observed for five consecutive epochs, and an "EarlyStopping" callback to terminate training if no improvement in validation loss was noted for 200 epochs, restoring the weights from the best epoch.

Performance Metrics

The model's performance was evaluated using validation accuracy and Categorical Cross-Entropy loss (target labels where one-hot encoded). Categorical Cross-Entropy Loss measures the performance of a classification model whose output is a probability value between 0 and 1. Categorical Cross-Entropy compares the distribution of the predictions with the true distribution of the labels and is calculated using the following formula for each label t and prediction y :

$$Loss = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M t_{ij} \log(y_{ij})$$

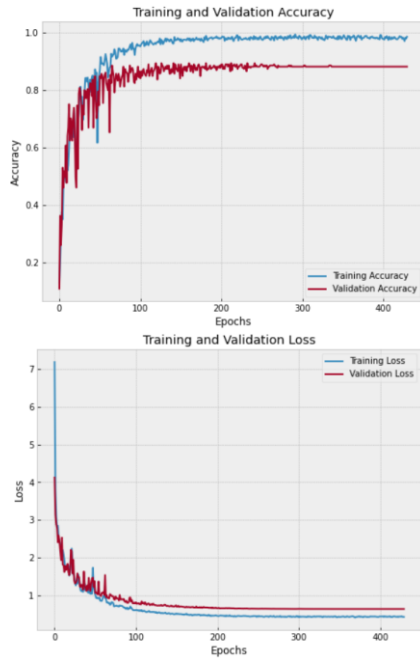
Validation Accuracy is a metric that measures the proportion of correct predictions (both true positives and true negatives) over all predictions made. It is defined as:

$$Accuracy = \frac{\# \text{ of correct predictions}}{\text{Total \# of predictions}}$$

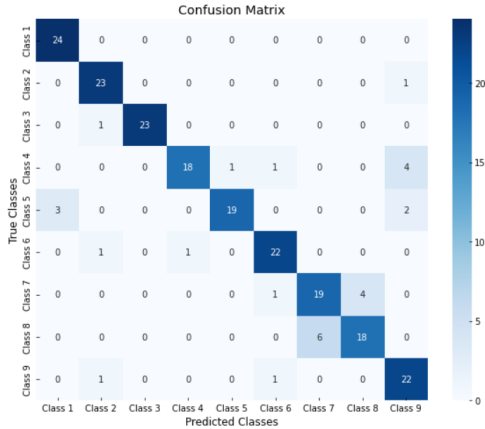
Experimental Findings

Various experiments were conducted adjusting parameters such as the number of layers, number of neurons per layer, learning rate, activations, kernel-regularizers, kernel-initializers, and dropout. It was found that having 2-3 layers with decreasing neurons in each layer was enough for the model to learn the dataset accurately. Additionally, using an adaptive learning rate in conjunction with the "ReduceLROnPlateau" conditional, resulted in better exploration of the loss landscape and ultimately a more accurate model. The best performing activation function for our purpose was the Exponential Linear Unit (ELU).

The model showed a marked improvement in recognizing ASL signs, as evidenced by the rising trend in training and validation accuracy over epochs. Our final validation accuracy was 90%.



Fine-tuning improved the model's performance, indicating that subtle, sign-specific features were captured when the base layers adjusted to the new domain. The experimental results underscore the importance of balancing the training phases to maximize the transfer of learning from a general domain (ImageNet) to a specific task (ASL signs classification).



The final confusion matrix is shown above. The model seemed to struggle particularly with classes H and G. Overall, however it seemed to approach decent generalization.

IV. CONCLUSIONS

This project demonstrated the application of transfer learning using the VGG19 convolutional neural network to classify

American Sign Language (ASL) signs, yielding several important insights. Firstly, the efficiency of transfer learning was evident, as utilizing the pre-trained VGG19 model not only accelerated the training process but also enhanced accuracy on the specialized ASL dataset. This underscores the utility of leveraging existing model's capabilities, which are adaptable to specialized tasks beyond their original training.

The phased training approach, starting with the freezing of the VGG19 base layers followed by gradual unfreezing for fine-tuning, was particularly effective. This method allowed the model to adapt incrementally to the unique characteristics of ASL signs, as reflected in the progressive improvement in validation accuracy and loss metrics. This strategy ensures that the model adjusts more finely to new data, optimizing performance.

Additionally, the project addressed the challenge of detecting unknown classes by implementing a confidence threshold-based method. This approach proved essential for enhancing the model's robustness, ensuring reliable performance even when confronted with signs absent from the training dataset. It highlights the broader applicability and adaptability of the model in real-world scenarios, where unexpected data is commonplace.

Finally, the use of dimensionality reduction techniques such as PCA, LDA, MDS, and ISOMAP provided insights into the data structure, aiding in the visual assessment of class separability and informing further model adjustments. These visualizations confirmed the complexity of the data's manifold and the applicability of the chosen techniques in capturing this complexity.

In conclusion, the project not only reinforced the potential of transfer learning in deploying deep learning models for specialized tasks but also demonstrated the importance of methodical approaches in model training and the handling of new or unknown data categories.

REFERENCES

- [1] Simonyan, K., & Zisserman, A. (2014). Very Deep Convolutional Networks for Large-Scale Image Recognition. arXiv:1409.1556.
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- [3] Garcia, N., & Bruna, J. (2019). Few-Shot Learning with Graph Neural Networks. ICLR.